SVM - Practical Implementation

Virtual Labs - Dayalbagh Educational Institute

Step 1: Importing Libraries

Importing Libraries

```
# Numerical and data handling libraries
import numpy as np
import pandas as pd

# Visualization libraries
import matplotlib.pyplot as plt
import seaborn as sns
from mlxtend.plotting import plot_decision_regions

print("Loaded libraries")
```

Output:
```
Loaded libraries
```

Dataset utilities

```
from sklearn.datasets import load_wine, make_moons
# Preprocessing and model utilities
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler
# Models
from sklearn.svm import SVC
# Evaluation metrics
from sklearn.metrics import (
    accuracy_score,
    precision_score,
    recall_score,
    f1_score,
    classification_report,
    confusion_matrix
)

print("Loaded Sklearn libraries")
```

Output:
```
Loaded Sklearn libraries
```

Step 2: Loading Dataset

Loading and Reading Data

```
wine = load_wine()
wine_df = pd.DataFrame(wine.data, columns=wine.feature_names)

wine_df["target"] = wine.target
wine_df
```

Output:

```
alcohol  malic_acid  ash   alcalinity_of_ash  magnesium  total_phenols  flavanoids
nonflavanoid_pheno  proanthocyanins  color_intensity  hue   od280/od315_of_dil  proline  target
0    14.23    1.71        2.43  15.6                  127.0      2.80            3.06         0.28
2.29            5.64             1.04  3.92                  1065.0   0
1    13.20    1.78        2.14  11.2                  100.0      2.65            2.76         0.26
1.28            4.38             1.05  3.40                  1050.0   0
2    13.16    2.36        2.67  18.6                  101.0      2.80            3.24         0.30
2.81            5.68             1.03  3.17                  1185.0   0
3    14.37    1.95        2.50  16.8                  113.0      3.85            3.49         0.24
2.18            7.80             0.86  3.45                  1480.0   0
4    13.24    2.59        2.87  21.0                  118.0      2.80            2.69         0.39
1.82            4.32             1.04  2.93                  735.0    0
...
173  13.71    5.65        2.45  20.5                  95.0       1.68            0.61         0.52
1.06            7.70             0.64  1.74                  740.0    2
174  13.40    3.91        2.48  23.0                  102.0      1.80            0.75         0.43
1.41            7.30             0.70  1.56                  750.0    2
175  13.27    4.28        2.26  20.0                  120.0      1.59            0.69         0.43
1.35            10.20            0.59  1.56                  835.0    2
176  13.17    2.59        2.37  20.0                  120.0      1.65            0.68         0.53
1.46            9.30             0.60  1.62                  840.0    2
177  14.13    4.10        2.74  24.5                  96.0       2.05            0.76         0.56
1.35            9.20             0.61  1.60                  560.0    2
178 rows x 14 columns
```

Display Wine dataset info

```
wine_df.info()
```

Output:

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 178 entries, 0 to 177
Data columns (total 14 columns):
#   Column                        Non-Null Count  Dtype
---  ------                        --------------  -----
0   alcohol                       178 non-null    float64
1   malic_acid                    178 non-null    float64
2   ash                           178 non-null    float64
3   alcalinity_of_ash             178 non-null    float64
4   magnesium                     178 non-null    float64
5   total_phenols                 178 non-null    float64
6   flavanoids                    178 non-null    float64
7   nonflavanoid_phenols          178 non-null    float64
8   proanthocyanins               178 non-null    float64
9   color_intensity               178 non-null    float64
10  hue                           178 non-null    float64
11  od280/od315_of_diluted_wines  178 non-null    float64
12  proline                       178 non-null    float64
13  target                        178 non-null    int32
dtypes: float64(13), int32(1)
memory usage: 19.0 KB
```

Generate two-moons dataset

```
X_moons, y_moons = make_moons(n_samples=700, noise=0.15, random_state=42)
moons_df = pd.DataFrame(X_moons, columns=["Feature_1", "Feature_2"])
moons_df["target"] = y_moons
moons_df
```

Output:

```
Feature_1  Feature_2  target
0    -0.036777  1.017428   0
1    0.991181   -0.542893  1
2    0.135605   0.095587   1
3    0.380490   0.882966   0
4    -0.795374  0.193136   0
...
695  1.147087   0.520498   0
696  0.200663   0.598349   0
697  -0.384952  0.547244   0
698  0.452570   -0.174473  1
699  0.534527   0.910810   0
700 rows x 3 columns
```

Step 3: Data Analysis

Data Analysis

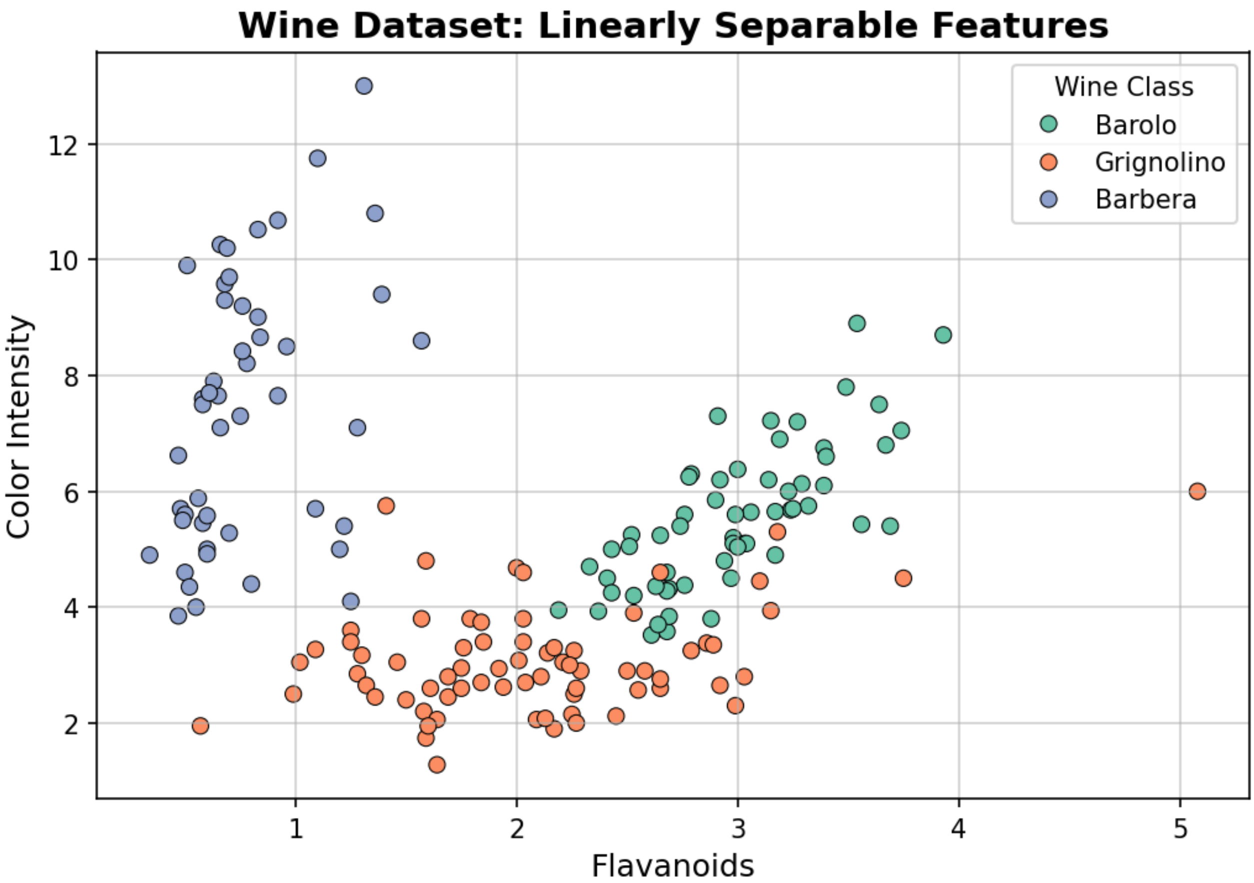
```
wine_df = pd.concat(
    [wine_df[["flavanoids", "color_intensity"]], wine_df["target"]],
    axis=1
)

wine_target_names = {
    0: "Barolo",
    1: "Grignolino",
    2: "Barbera"
}

wine_df["class_name"] = wine_df["target"].map(wine_target_names)
wine_df[['target', 'class_name']].value_counts()
```

Output:

```
target  class_name
1       Grignolino    71
0       Barolo        59
2       Barbera       48
Name: count, dtype: int64
```

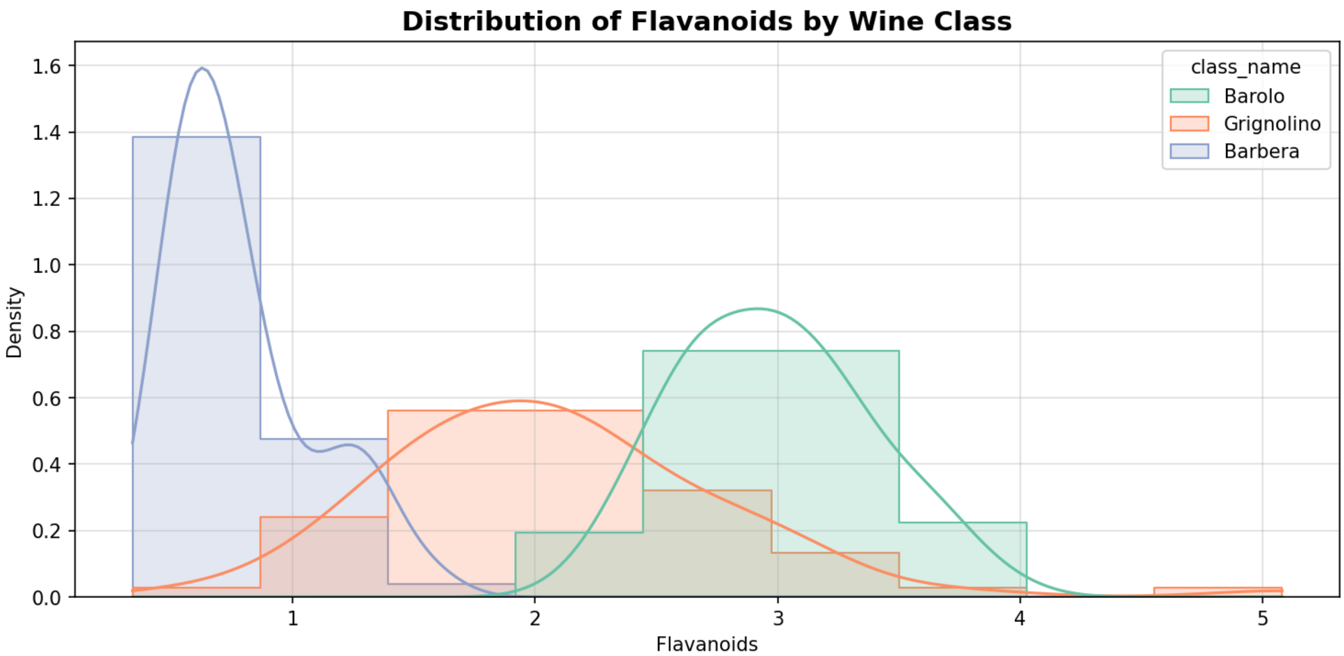
Visualizing Wine Dataset

```
plt.figure(figsize=(7, 5))
sns.scatterplot(
    data=wine_df,
    x="flavanoids",
    y="color_intensity",
    hue="class_name",
    palette="Set2",
    s=40,
    edgecolor="black"
)
plt.title("Wine Dataset: Linearly Separable Features", fontsize=14, fontweight="bold")
plt.xlabel("Flavanoids", fontsize=12)
plt.ylabel("Color Intensity", fontsize=12)
plt.legend(title="Wine Class")
plt.grid(alpha=0.6)
plt.show()
```



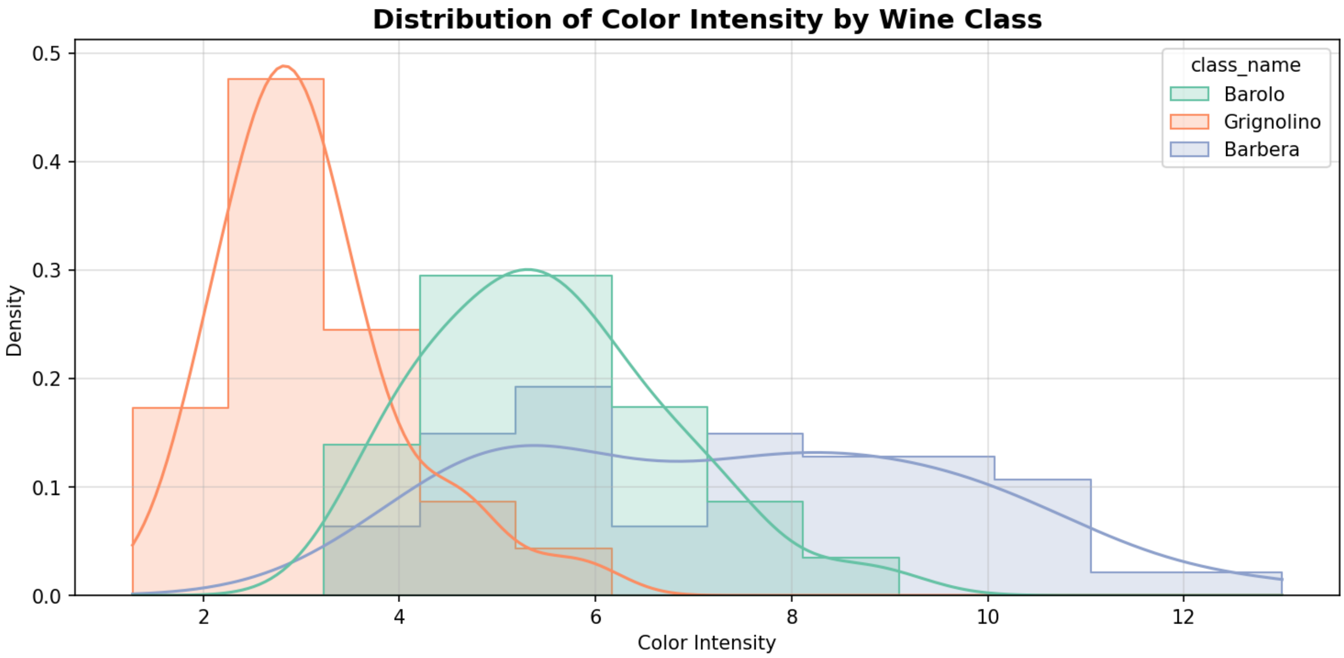
Histogram of Flavanoids

```
plt.figure(figsize=(10, 5))
sns.histplot(
    data=wine_df,
    x="flavanoids",
    hue="class_name",
    kde=True,
    palette="Set2",
    element="step",
    stat="density",
    common_norm=False
)
plt.title("Distribution of Flavanoids by Wine Class", fontsize=14, fontweight="bold")
plt.xlabel("Flavanoids")
plt.ylabel("Density")
plt.grid(alpha=0.4)
plt.show()
```



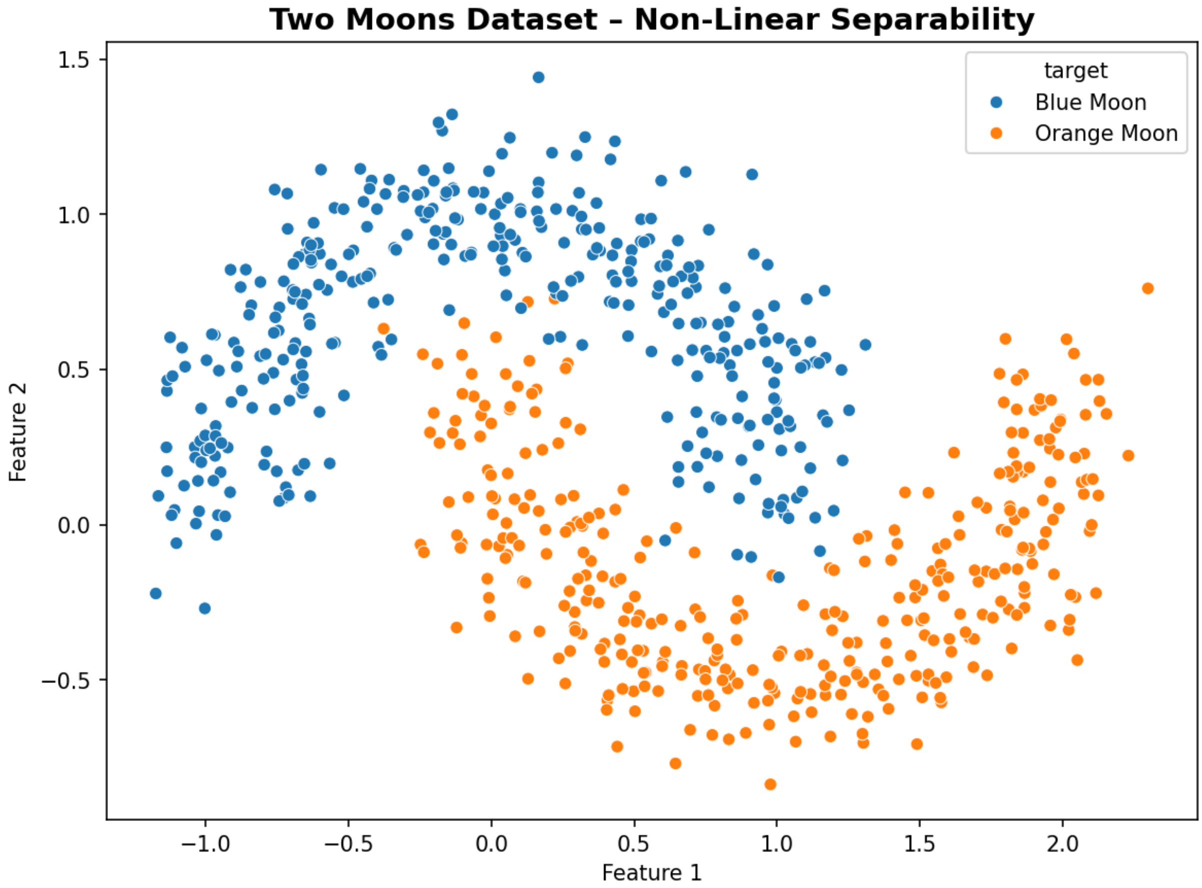
Histogram of Color Intensity

```
plt.figure(figsize=(10, 5))
sns.histplot(
    data=wine_df,
    x="color_intensity",
    hue="class_name",
    kde=True,
    palette="Set2",
    element="step",
    stat="density",
    common_norm=False
)
plt.title("Distribution of Color Intensity by Wine Class", fontsize=14, fontweight="bold")
plt.xlabel("Color Intensity")
plt.ylabel("Density")
plt.grid(alpha=0.4)
plt.show()
```



Visualizing Two Moons Dataset

```
plt.figure(figsize=(8,6))
sns.scatterplot(
    x=moons_df["Feature_1"],
    y=moons_df["Feature_2"],
    hue=moons_df["target"].map({0: "Blue Moon", 1: "Orange Moon"}),
    palette=["#1f77b4", "#ff7f0e"]
)
plt.title("Two Moons Dataset - Non-Linear Separability", fontsize=14, fontweight='bold')
plt.xlabel("Feature 1")
plt.ylabel("Feature 2")
plt.show()
```



Step 4: Data Preprocessing

 # Data Preprocessing

```
X_wine = wine_df[["flavanoids", "color_intensity"]]
y_wine = wine_df["target"]

Xw_train, Xw_test, yw_train, yw_test = train_test_split(
    X_wine, y_wine, test_size=0.2, random_state=42, stratify=y_wine
)
Xw_train.shape, Xw_test.shape, yw_train.shape, yw_test.shape
```

Output:

```
((142, 2), (36, 2), (142,), (36,))
```

 # Train/Test split for Two Moons dataset

```
Xm_train, Xm_test, ym_train, ym_test = train_test_split(
    X_moons, y_moons, test_size=0.2, random_state=42, stratify=y_moons
)
Xm_train.shape, Xm_test.shape, ym_train.shape, ym_test.shape
```

Output:

```
((560, 2), (140, 2), (560,), (140,))
```

Feature Scaling

```
scaler = StandardScaler()
scaler
```

Output:
```
StandardScaler()
```

Scale Wine training and test data

```
Xw_train_scaled = scaler.fit_transform(Xw_train)
Xw_test_scaled = scaler.transform(Xw_test)

print("Wine data scaled successfully!")
```

Output:
```
Wine data scaled successfully!
```

View original Wine training data (before scaling)

```
Xw_train[:5]
```

Output:
```
flavanoids  color_intensity
10   2.81        5.40
149  3.17        5.75
90   3.00        5.50
59   2.91        5.60
147  0.66        9.39
```

View scaled Wine training data (after scaling)

```
Xw_train_scaled[:5]
```

Output:
```
array([[ 0.73229212, -0.16746725],
[ 1.33318146,  0.30530313],
[ 1.006382  , -0.081509  ],
[ 0.81662747,  0.262324  ],
[-1.29175618,  1.47433535]])
```

Scale Two Moons training and test data

```
Xm_train_scaled = scaler.fit_transform(Xm_train)
Xm_test_scaled = scaler.transform(Xm_test)

print("Two Moons data scaled successfully!")
```

Output:
```
Two Moons data scaled successfully!
```

View original Two Moons training data (before scaling)

```
Xm_train[:5]
```

Output:
```
array([[ 1.3723603 , -0.55081882],
[ 0.63613117,  0.78345978],
[-0.43531115,  0.96004336],
[ 0.41718644,  1.17721089],
[-1.13531808,  0.17220389]])
```

 # View scaled Two Moons training data (after scaling)

```
Xm_train_scaled[:5]
```

Output:
```
array([[ 1.01466867, -1.57697415],
[ 0.17320777,  1.01862443],
[-1.05137949,  1.36213595],
[-0.07703149,  1.78459625],
[-1.85144083, -0.17046376]])
```

Step 5: Model Training

 # Model Training

```
# Traing linear Kernel SVM in wine dataset with 2 features
svm_linear_wine = SVC(kernel="linear")
svm_linear_wine.fit(Xw_train_scaled, yw_train)
svm_linear_wine
```

Output:
```
SVC(kernel='linear')
```

 # Training SVMs with linear and RBF kernels on two-moons toy dataset

```
svm_linear_moon = SVC(kernel="linear")
svm_rbf_moon = SVC(kernel="rbf")

svm_linear_moon.fit(Xm_train_scaled, ym_train)
svm_rbf_moon.fit(Xm_train_scaled, ym_train)
print("Model trained")
```

Output:
```
Model trained
```

Step 6: Model Evaluation

 # Model Evaluation

```
# Wine dataset case
wine_preds = svm_linear_wine.predict(Xw_test_scaled)
wine_preds
```

Output:
```
array([0, 2, 0, 0, 1, 0, 0, 0, 1, 2, 1, 2, 0, 1, 0, 1, 1, 0, 1, 1, 1, 1,
0, 0, 0, 1, 0, 2, 1, 2, 0, 2, 1, 2, 2, 2])
```

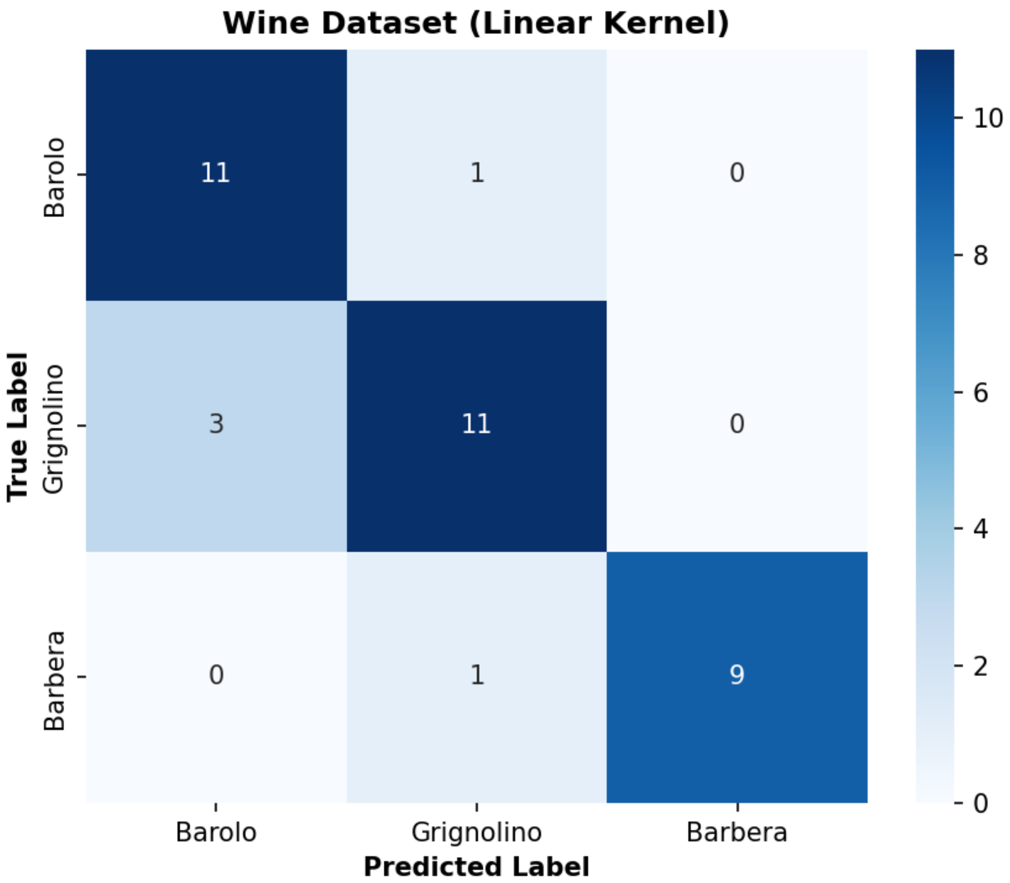
Wine dataset - Classification metrics

```
print("Accuracy:", accuracy_score(yw_test, wine_preds))
print("Precision:", precision_score(yw_test, wine_preds, average="macro"))
print("Recall:", recall_score(yw_test, wine_preds, average="macro"))
print("F1 Score:", f1_score(yw_test, wine_preds, average="macro"))

print("\\nClassification Report:\\n")
print(classification_report(yw_test, wine_preds))
```

Output:
```
Accuracy: 0.8611111111111112
Precision: 0.8772893772893773
Recall: 0.8674603174603175
F1 Score: 0.8694456940070975
Classification Report:
precision    recall  f1-score   support
0       0.79      0.92      0.85        12
1       0.85      0.79      0.81        14
2       1.00      0.90      0.95        10
accuracy                           0.86        36
macro avg       0.88      0.87      0.87        36
weighted avg       0.87      0.86      0.86        36
```

Confusion Matrix

```
labels = list(wine_target_names.keys())
class_names = list(wine_target_names.values())

plt.figure(figsize=(6, 5))
sns.heatmap(
    confusion_matrix(yw_test, wine_preds, labels=labels),
    annot=True,
    fmt="d",
    cmap="Blues",
    xticklabels=class_names,
    yticklabels=class_names
)

plt.title("Wine Dataset (Linear Kernel)", fontweight='bold')
plt.xlabel("Predicted Label", fontweight='bold')
plt.ylabel("True Label", fontweight='bold')
plt.tight_layout()
plt.show()
```



Two moon dataset case

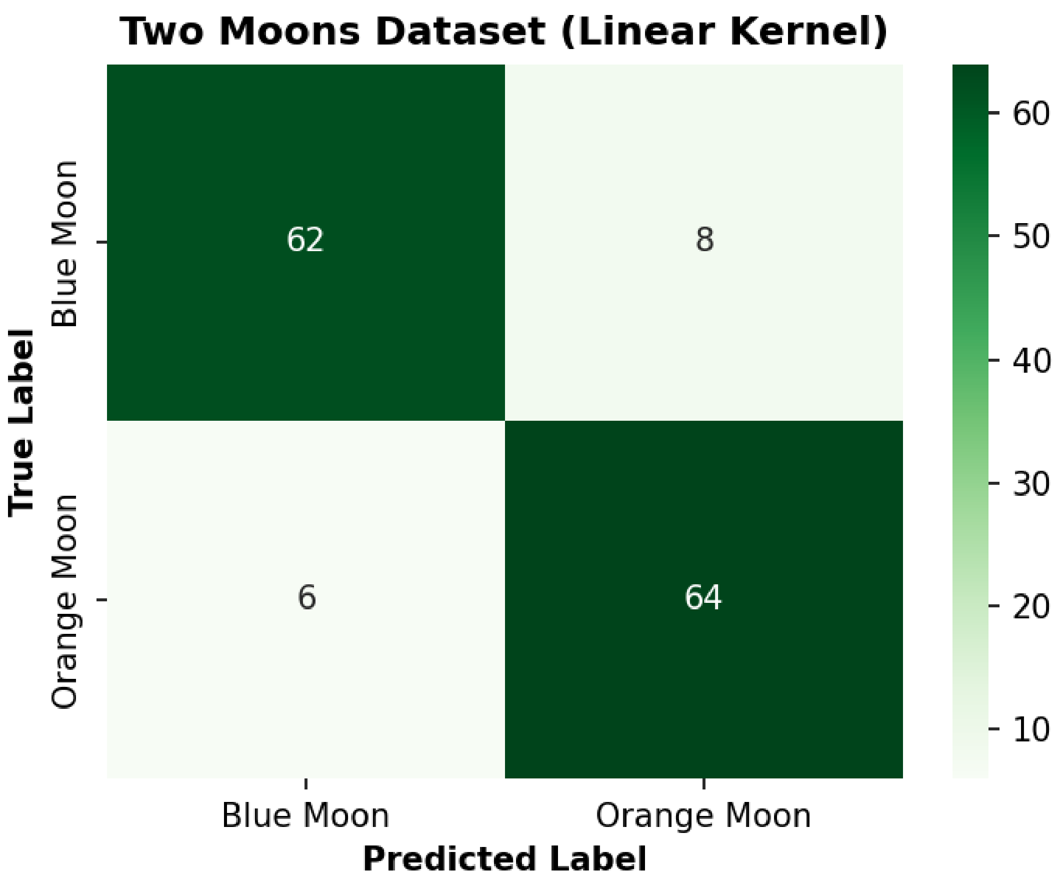
```
moon_linear_preds = svm_linear_moon.predict(Xm_test_scaled)
moon_linear_preds
```

Output:

```
array([0, 0, 1, 1, 1, 1, 1, 1, 1, 0, 1, 1, 1, 1, 1, 0, 0, 0, 0, 0, 0, 0,
0, 1, 1, 0, 1, 0, 1, 1, 0, 1, 1, 0, 0, 1, 1, 1, 0, 0, 1, 0, 1, 1,
1, 1, 1, 1, 1, 1, 0, 0, 0, 0, 1, 1, 0, 0, 1, 0, 1, 0, 1, 0, 0, 1,
0, 1, 0, 1, 1, 0, 0, 1, 1, 0, 1, 0, 0, 0, 1, 0, 0, 0, 1, 1, 0, 0,
1, 0, 0, 1, 1, 0, 0, 0, 0, 1, 1, 1, 1, 1, 1, 1, 0, 0, 0, 0, 1, 0,
1, 0, 0, 1, 1, 1, 1, 1, 0, 1, 0, 0, 1, 1, 0, 1, 1, 0, 0, 0, 0, 0,
0, 0, 1, 0, 0, 1, 1, 1])
```

Confusion Matrix

```
class_names = ['Blue Moon', 'Orange Moon']

plt.figure(figsize=(5, 4))
sns.heatmap(
    confusion_matrix(ym_test, moon_linear_preds),
    annot=True,
    fmt="d",
    cmap="Greens",
    xticklabels=class_names,
    yticklabels=class_names
)

plt.title("Two Moons Dataset (Linear Kernel)", fontweight='bold')
plt.xlabel("Predicted Label", fontweight='bold')
plt.ylabel("True Label", fontweight='bold')
plt.tight_layout()
plt.show()
```



Two Moons - Linear kernel classification metrics

```
print("Accuracy:", accuracy_score(ym_test, moon_linear_preds))
print("Precision:", precision_score(ym_test, moon_linear_preds, average="macro"))
print("Recall:", recall_score(ym_test, moon_linear_preds, average="macro"))
print("F1 Score:", f1_score(ym_test, moon_linear_preds, average="macro"))

print("\\nClassification Report:\\n")
print(classification_report(ym_test, moon_linear_preds))
```

Output:

```
Accuracy: 0.9
Precision: 0.9003267973856208
Recall: 0.8999999999999999
F1 Score: 0.8999795876709533
Classification Report:
precision    recall  f1-score   support
0       0.91      0.89      0.90        70
1       0.89      0.91      0.90        70
accuracy                           0.90       140
macro avg       0.90      0.90      0.90       140
weighted avg       0.90      0.90      0.90       140
```

Two moon dataset case

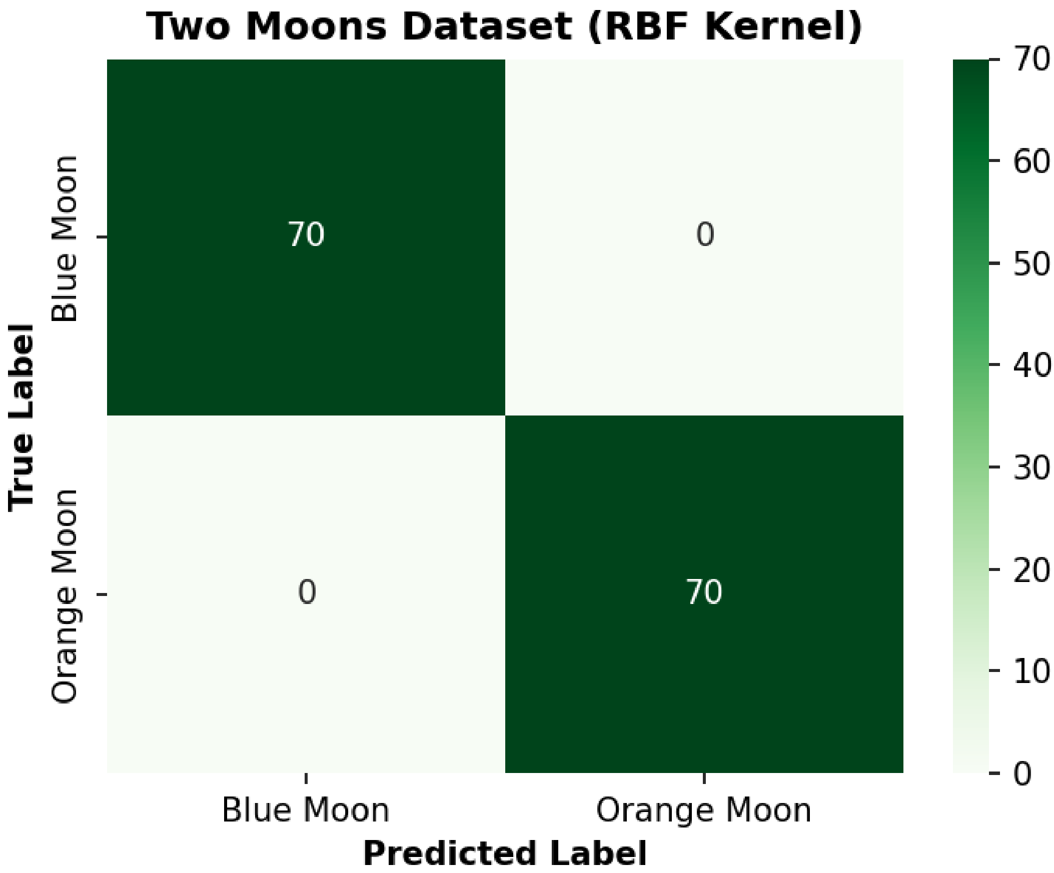
```
moon_rbf_preds = svm_rbf_moon.predict(Xm_test_scaled)
moon_rbf_preds
```

Output:

```
array([0, 0, 1, 0, 1, 1, 1, 1, 1, 0, 1, 1, 1, 1, 1, 0, 0, 0, 0, 0, 0, 0,
0, 0, 1, 0, 1, 1, 1, 1, 0, 1, 1, 0, 0, 1, 1, 1, 0, 0, 1, 0, 1, 1,
1, 1, 1, 1, 1, 1, 0, 0, 0, 0, 0, 1, 0, 0, 0, 0, 1, 0, 1, 0, 0, 1,
0, 1, 1, 1, 1, 0, 0, 1, 1, 0, 1, 0, 0, 0, 0, 0, 0, 0, 1, 1, 0, 0,
1, 0, 0, 1, 1, 0, 0, 0, 0, 1, 1, 1, 1, 1, 1, 1, 0, 0, 0, 0, 1, 0,
1, 0, 0, 1, 0, 1, 1, 1, 0, 0, 0, 0, 1, 1, 0, 1, 1, 1, 0, 0, 0, 0,
1, 1, 1, 0, 1, 1, 0, 1])
```

Two Moons - RBF Kernel Confusion Matrix

```
class_names = ['Blue Moon', 'Orange Moon']

plt.figure(figsize=(5, 4))
sns.heatmap(
    confusion_matrix(ym_test, moon_rbf_preds),
    annot=True,
    fmt="d",
    cmap="Greens",
    xticklabels=class_names,
    yticklabels=class_names
)

plt.title("Two Moons Dataset (RBF Kernel)", fontweight='bold')
plt.xlabel("Predicted Label", fontweight='bold')
plt.ylabel("True Label", fontweight='bold')
plt.tight_layout()
plt.show()
```



 # Two Moons - RBF kernel classification metrics (Perfect accuracy!)

```
print("Accuracy:", accuracy_score(ym_test, moon_rbf_preds))
print("Precision:", precision_score(ym_test, moon_rbf_preds, average="macro"))
print("Recall:", recall_score(ym_test, moon_rbf_preds, average="macro"))
print("F1 Score:", f1_score(ym_test, moon_rbf_preds, average="macro"))

print("\\nClassification Report:\\n")
print(classification_report(ym_test, moon_rbf_preds))
```

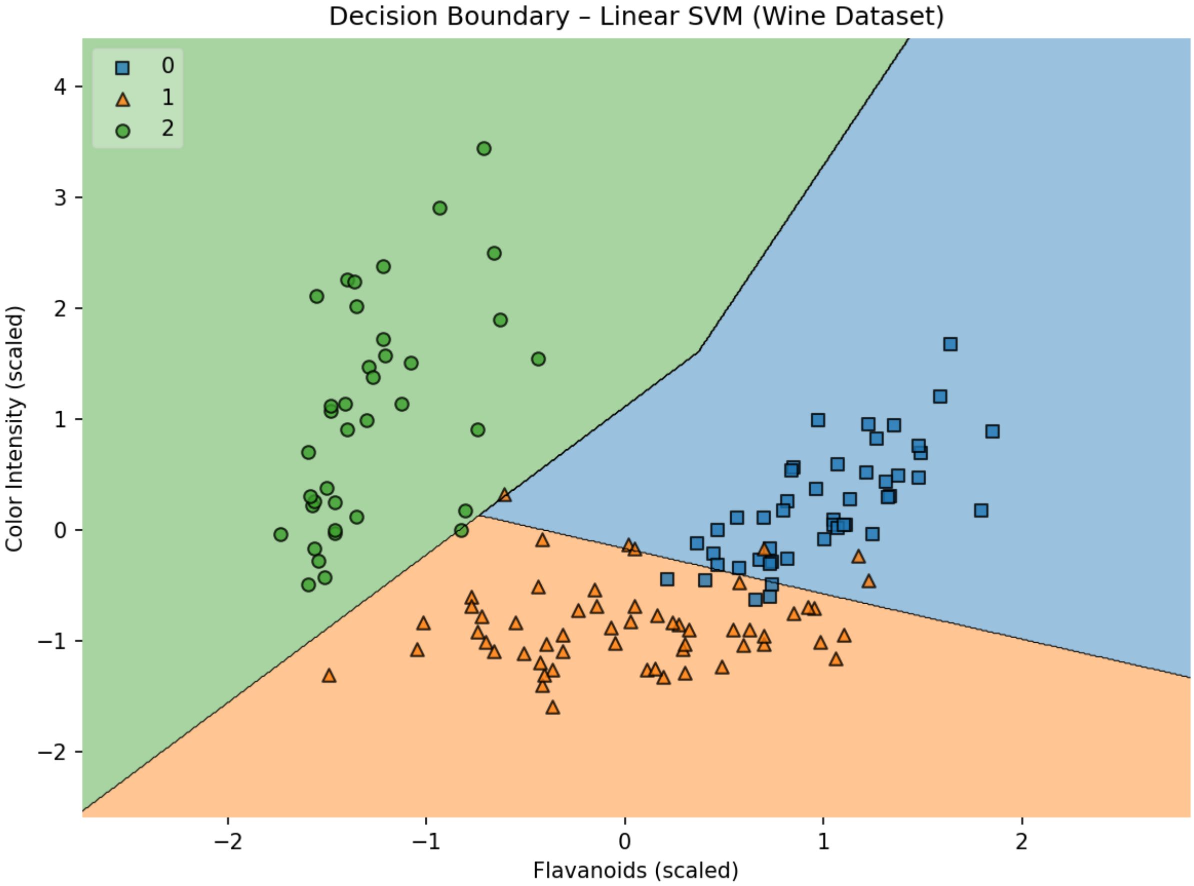
Output:

```
Accuracy: 1.0
Precision: 1.0
Recall: 1.0
F1 Score: 1.0
Classification Report:
precision    recall  f1-score   support
0        1.00      1.00      1.00        70
1        1.00      1.00      1.00        70
accuracy                          1.00       140
macro avg       1.00      1.00      1.00       140
weighted avg       1.00      1.00      1.00       140
```

Step 7: Model Simulation

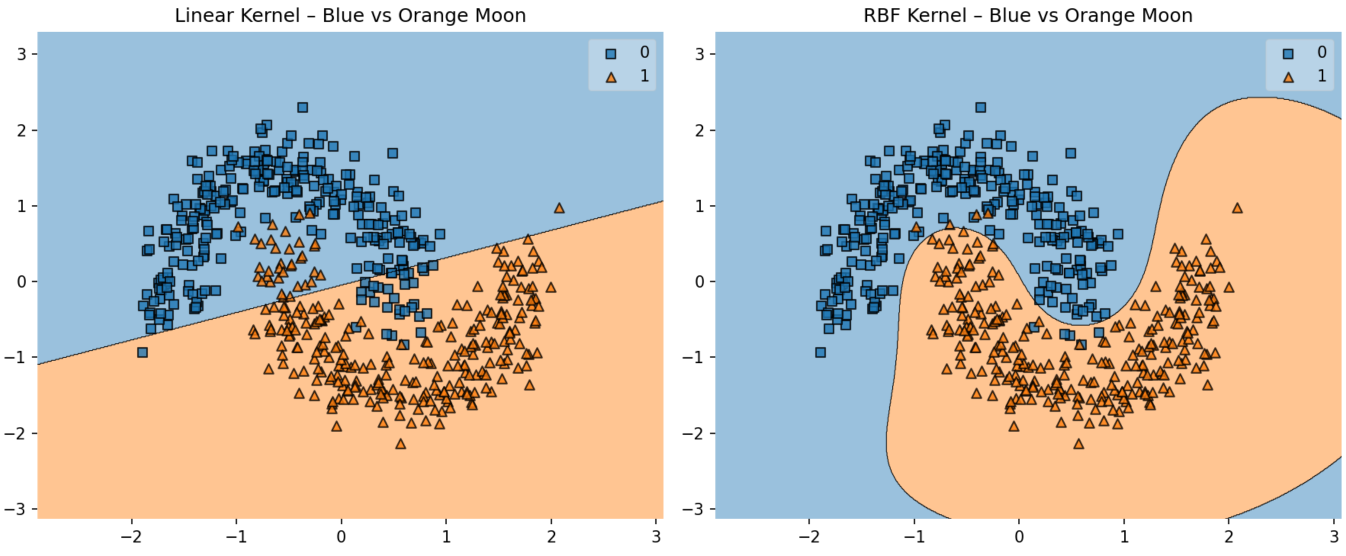
 # Model testing

```
# Decision Boundaries - Wine
plot_decision_regions(
    Xw_train_scaled,
    yw_train.values,
    clf=svm_linear_wine,
    legend=2
)
plt.title("Decision Boundary - Linear SVM (Wine Dataset)")
plt.xlabel("Flavanoids (scaled)")
plt.ylabel("Color Intensity (scaled)")
plt.show()
```



 # Decision Boundaries - Two Moons

```
plt.figure(figsize=(12,5))

plt.subplot(1,2,1)
plot_decision_regions(Xm_train_scaled, ym_train, clf=svm_linear_moon)
plt.title("Linear Kernel - Blue vs Orange Moon")

plt.subplot(1,2,2)
plot_decision_regions(Xm_train_scaled, ym_train, clf=svm_rbf_moon)
plt.title("RBF Kernel - Blue vs Orange Moon")

plt.show()
```


Linear Kernel – Blue vs Orange Moon
RBF Kernel – Blue vs Orange Moon
0
1
3
2
1
0
−1
−2
−3
−2
−1
0
1
2
3